Mathematics for 3D Graphics

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March 25, 2026

Overview

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Mathematical Objects Used for 3D Graphics

- Vectors
- Matrices
- Homogenous coordinates
- Bézier curves
- Splines
- Quaternions

Bézier curves

Bézier curves

- Parametric curve $B(t)$ with $t \in [0, 1]$
- Defined by control points $P_0, \ldots, P_n$
- Interpolates endpoints: $B(0) = P_0$, $B(1) = P_n$
- Control points shape the curve
- Common in fonts, CAD, animation, path planning

Bernstein form

$$B(t) = \sum_{i=0}^{n} \binom{n}{i} (1-t)^{n-i} t^i P_i$$

- Basis functions are Bernstein polynomials
- Coefficients are nonnegative for $t \in [0, 1]$
- Coefficients sum to 1 (convex combination)

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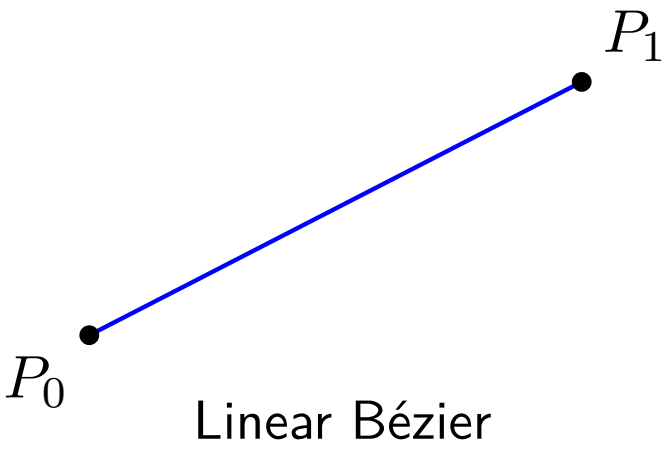
Degree 1 and 2 Bézier equations

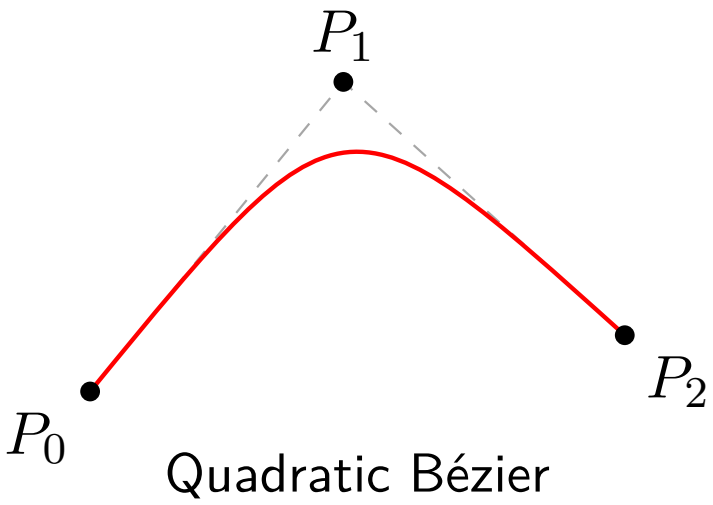
Degree 1 (linear):

$$B_1(t) = (1-t)P_0 + tP_1$$

Degree 2 (quadratic):

$$B_2(t) = (1-t)^2 P_0 + 2(1-t)tP_1 + t^2 P_2$$

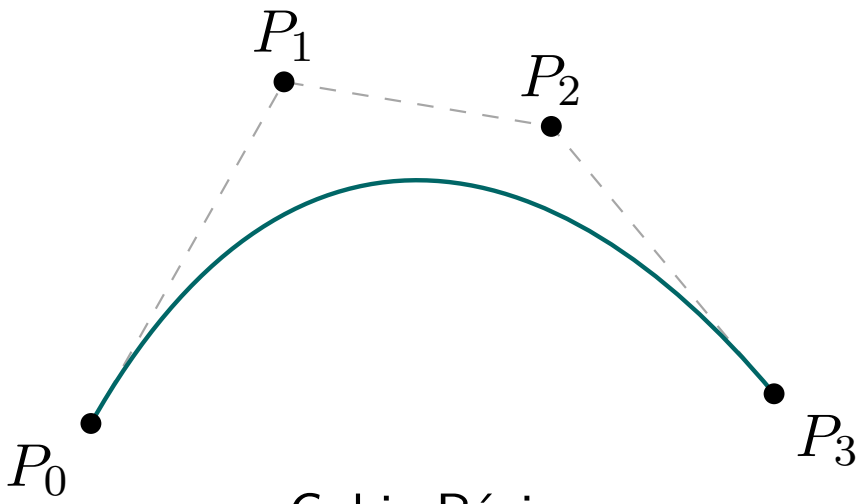




Degree 3 Bézier equation

$$B_3(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t)t^2 P_2 + t^3 P_3$$

- Cubic is the most common in graphics APIs
- 4 control points, smooth and expressive
- Segment chaining controls continuity
- Parameterized by $t \in [0, 1]$



Cubic Bézier

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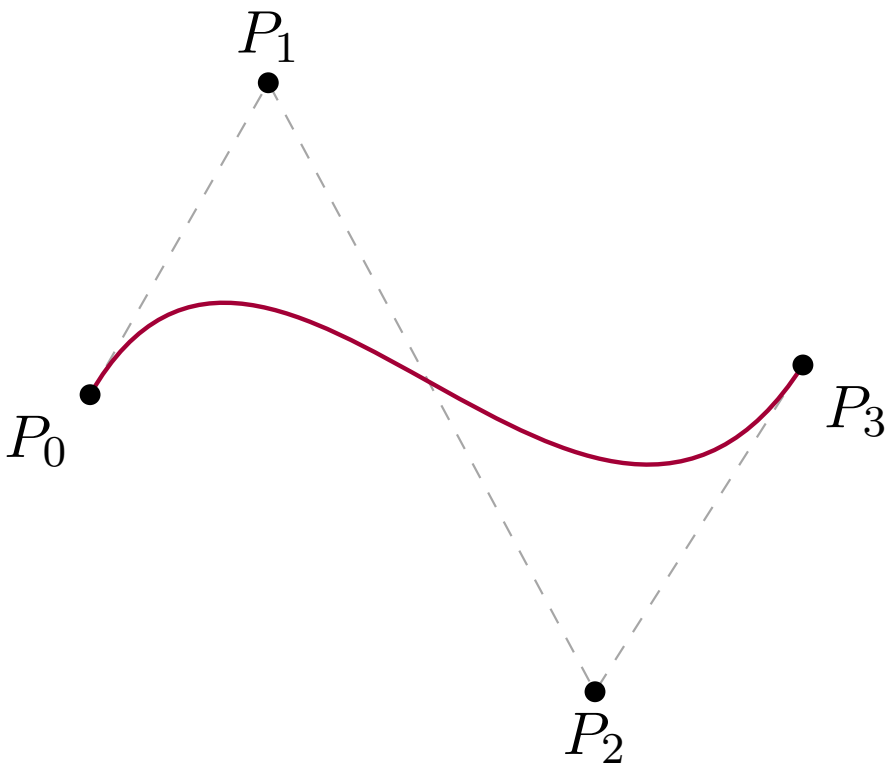
Degree 3 Bézier curves continued



Cubic Bézier S-bend

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Example problem in 2D

Given:

$$P_0 = (0,0),\ P_1 = (2,2),\ P_2 = (4,0),\ t = 0.5$$

Find $B_2(0.5)$ **for**

$$B_2(t) = (1-t)^2 P_0 + 2(1-t)tP_1 + t^2 P_2$$

$$B_2(0.5) = 0.25P_0 + 0.5P_1 + 0.25P_2$$

$$= 0.25(0,0) + 0.5(2,2) + 0.25(4,0) = (2,1)$$

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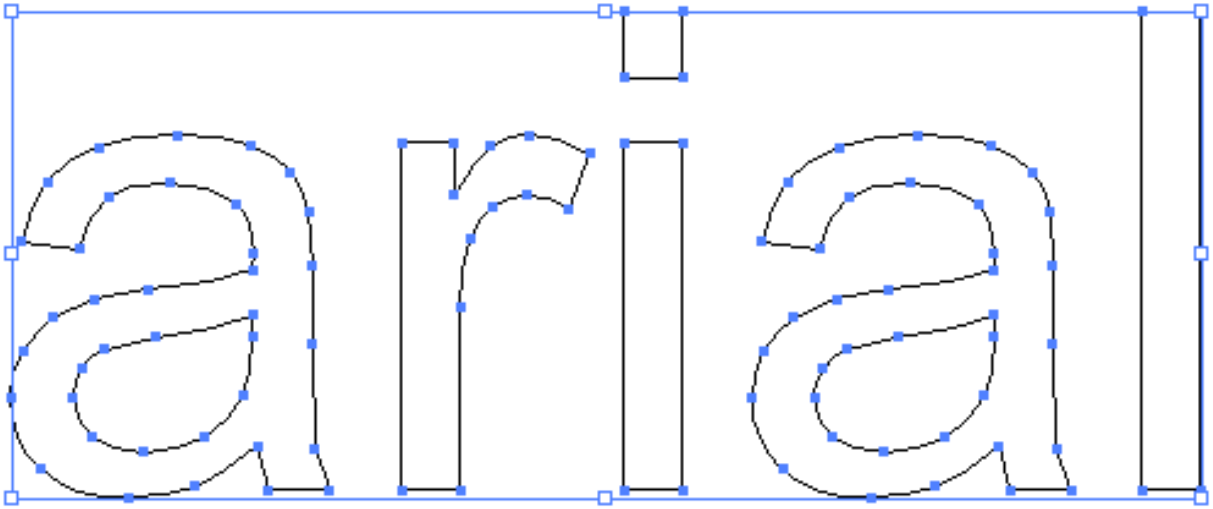
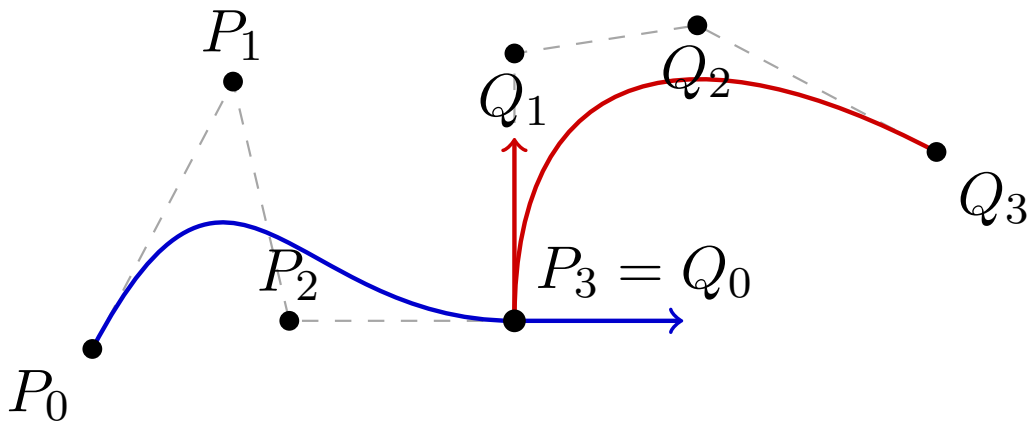
Applications of Bézier curves

- Fonts
- CAD
- Animation
- Path planning



Path Planning with Bézier curves

- Derivative is known analytically
- 2nd derivative may be discontinuous
- This may not be physically possible



Adjacent cubics with C^0 but not C^1 continuity

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Length of a Bézier curve

- Arclength of a Bézier curve is given by:

$$L = \int_0^1 \sqrt{B'(t)^2} dt$$

- There is no analytical solution for this integral
- Approximated numerically

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Example: Calculate Length

$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t)t^2 P_2 + t^3 P_3$$

$$P_0 = (0,0), P_1 = (1,1), P_2 = (2,1), P_3 = (3,0)$$

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Derivative

$$B'(t) = 3(1-t)^2(P_1 - P_0) + 6(1-t)t(P_2 - P_1) + 3t^2(P_3 - P_2)$$

$$P_1 - P_0 = (1,1), \quad P_2 - P_1 = (1,0), \quad P_3 - P_2 = (1,-1)$$

$$B'(t) = 3(1-t)^2(1,1) + 6(1-t)t(1,0) + 3t^2(1,-1)$$

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Simplification

$$x'(t) = 3\left[(1-t)^2 + 2(1-t)t + t^2\right] = 3$$

$$y'(t) = 3\left[(1-t)^2 - t^2\right] = 3(1-2t)$$

$$\|B'(t)\| = \sqrt{9 + 9(1-2t)^2} = 3\sqrt{1 + (1-2t)^2}$$

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Arc Length Integral

$$L = \int_0^1 \|B'(t)\|\, dt = \int_0^1 3\sqrt{1 + (1-2t)^2}\, dt$$

Substitute:

$$u = 1 - 2t, \quad dt = -\frac{1}{2} du$$

$$L = \frac{3}{2} \int_{-1}^1 \sqrt{1 + u^2}\, du$$

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Closed Form

$$\int \sqrt{1+u^2}\,du = \frac{1}{2}\left(u\sqrt{1+u^2} + \sinh^{-1}(u)\right)$$

$$\int_{-1}^{1} \sqrt{1+u^2}\,du = \sqrt{2} + \sinh^{-1}(1)$$

$$L = \frac{3}{2}\left(\sqrt{2} + \sinh^{-1}(1)\right)$$

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Numerical Value

$$\sqrt{2} \approx 1.4142, \quad \sinh^{-1}(1) = \ln(1+\sqrt{2}) \approx 0.88137$$

$$L \approx \frac{3}{2}(2.2956) \approx 3.4434$$

$$\boxed{L \approx 3.4434}$$

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Splines

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Splines

- Splines represent smooth curves in 3D space.
- Original splines were physical pieces of wood
- They are defined by a set of control points but changing one changes all
- Solved using $n + 1$ equations for $n + 1$ control points

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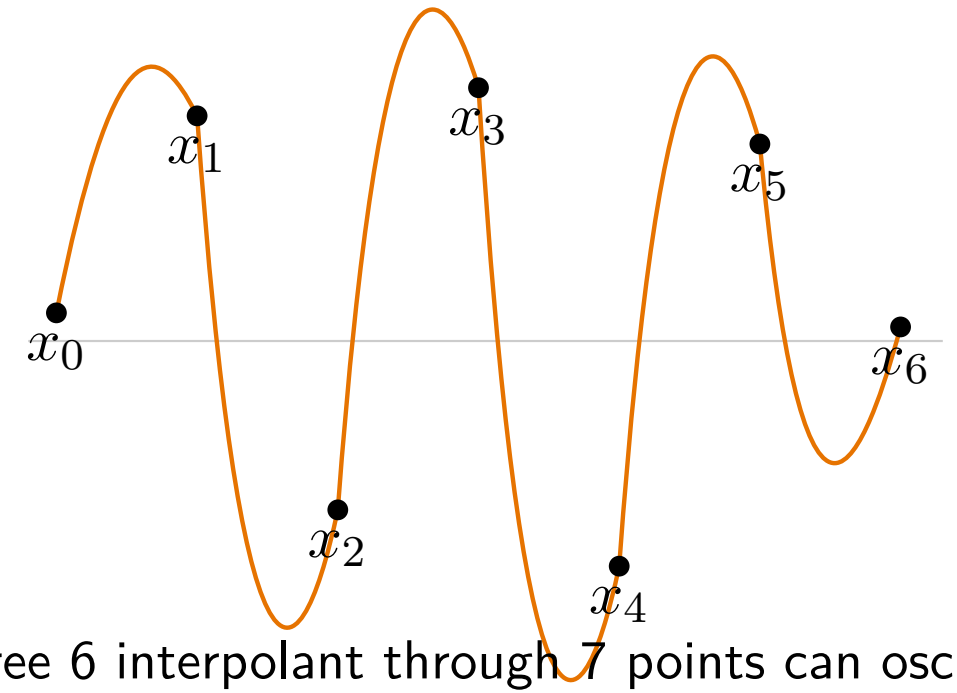
High-order Polynomials: Not smooth

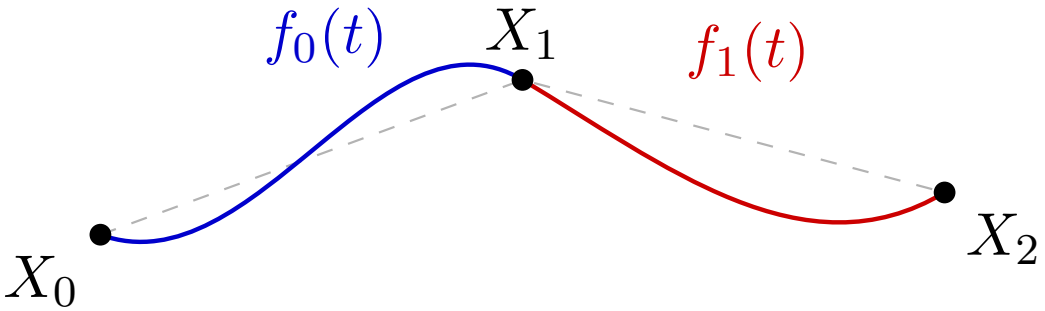
- A polynomial of degree n has $n+1$ coefficients.
- can interpolate $n+1$ points exactly.
- However, it will oscillate wildly between the points.



Degree 6 interpolant through 7 points can oscillate

Two cubic spline regions

- Two piecewise cubic functions
- Left segment uses $f_0(t)$, right segment uses $f_1(t)$
- Segments meet at X_1



Quaternions

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Quaternions

- Quaternions represent rotations in 3D space.
- Generalization of complex numbers.
- One real part and three imaginary parts.

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Quaternions

- The real value w represents the rotation angle.
- The vector (x, y, z) represents the rotation axis.
- $$q = w + xi + yj + zk$$

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Quaternion Operations

- Addition:

$$q_1 + q_2 = (w_1 + w_2) + (x_1 + x_2)i + (y_1 + y_2)j + (z_1 + z_2)k$$

- Multiplication:

$$\begin{aligned} q_1 \times q_2 = &(w_1 w_2 - x_1 x_2 - y_1 y_2 - z_1 z_2) \\ &+ (w_1 x_2 + x_1 w_2 + y_1 z_2 - z_1 y_2)i \\ &+ (w_1 y_2 - x_1 z_2 + y_1 w_2 + z_1 x_2)j \\ &+ (w_1 z_2 + x_1 y_2 - y_1 x_2 + z_1 w_2)k \end{aligned}$$

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Quaternion Operations Continued

- Conjugate:

$$q^* = w - xi - yj - zk$$

- Inverse:

$$q^{-1} = \frac{q^*}{|q|^2}$$

- Norm:

$$|q| = \sqrt{w^2 + x^2 + y^2 + z^2}$$

- Normalize:

$$\frac{q}{|q|} = \frac{q}{\sqrt{w^2 + x^2 + y^2 + z^2}}$$

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Quaternion Practice

- $$q_1 = (1, 0, 0, 0)$$
- $$q_2 = (0, 1, 0, 0)$$
- $$q_3 = (0, 0, 1, 0)$$
- $$q_4 = (0, 0, 0, 1)$$
- $$q_1 + q_2 = (1, 1, 0, 0)$$
- $$q_1 \times q_2 = (0, 0, 0, 1)$$
- $$q_1^* = (1, 0, 0, 0)$$

Rotate a Vector with a Quaternion

Given:

- Vector $v = (1, 0, 0)$
- Rotate 90° about the z-axis

Quaternion for rotation:

$$q = (\cos\tfrac{\theta}{2},\ 0,\ 0,\ \sin\tfrac{\theta}{2}) = \left(\tfrac{\sqrt{2}}{2},\ 0,\ 0,\ \tfrac{\sqrt{2}}{2}\right)$$

Treat vector as quaternion:

$$v_q = (0, 1, 0, 0)$$

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Quaternion Rotation Formula

Rotate using:

$$v' = q\, v_q\, q^{-1}$$

For unit quaternions:

$$q^{-1} = \bar{q} = (w, -x, -y, -z)$$

$$q^{-1} = \left(\frac{\sqrt{2}}{2},\ 0,\ 0,\ -\frac{\sqrt{2}}{2}\right)$$

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First Multiply: $q \cdot v_q$

$$q = \left(\frac{\sqrt{2}}{2}, 0, 0, \frac{\sqrt{2}}{2}\right), \quad v_q = (0, 1, 0, 0)$$

Quaternion multiplication gives:

$$qv_q = \left(0,\ \frac{\sqrt{2}}{2},\ \frac{\sqrt{2}}{2},\ 0\right)$$

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Second Multiply: $(qv_q) \cdot q^{-1}$

$$(0, \tfrac{\sqrt{2}}{2}, \tfrac{\sqrt{2}}{2}, 0) \cdot \left(\tfrac{\sqrt{2}}{2}, 0, 0, -\tfrac{\sqrt{2}}{2}\right) = (0, 0, 1, 0)$$

Resulting vector:

$$v' = (0, 1, 0)$$

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Result

- Original vector: $(1, 0, 0)$
- Rotated 90° about z
- Result: $(0, 1, 0)$
- Quaternion rotation matches expected geometry

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Useful Properties

- Only 4 parameters (compact vs. 9 for matrices)
- No gimbal lock
- Efficient composition (rotation = quaternion multiply)
- Numerically stable (easy to renormalize)
- Smooth interpolation (SLERP)
- Inverse = conjugate for unit quaternions
- Avoids drift compared to Euler angle integration
- Easy conversion to/from rotation matrices

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Goal

Given:

- Start position P_0, end position P_3
- Desired start/end tangents (via P_1, P_2)
- Start orientation q_0, end orientation q_1

Construct:

- Smooth translation (cubic Bézier)
- Smooth rotation (quaternion interpolation)

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Position via Cubic Bézier

$$B(t) = (1-t)^3 P_0 + 3(1-t)^2 t P_1 + 3(1-t)t^2 P_2 + t^3 P_3, \quad t \in [0,1]$$

- P_0 = start position
- P_3 = end position
- P_1, P_2 control entry/exit direction

$$\text{Position}(t) = B(t)$$

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Velocity (Direction of Motion)

$$B'(t) = 3(1-t)^2(P_1 - P_0) + 6(1-t)t(P_2 - P_1) + 3t^2(P_3 - P_2)$$

- Tangent gives instantaneous direction
- Can be used to align forward axis if desired

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Orientation via Quaternions

- q_0 = initial orientation
- q_1 = final orientation

Use spherical linear interpolation (SLERP):

$$q(t) = \frac{\sin((1-t)\theta)}{\sin\theta} q_0 + \frac{\sin(t\theta)}{\sin\theta} q_1$$

where:

$$\cos\theta = q_0 \cdot q_1$$

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Properties of SLERP

- Constant angular velocity
- Shortest path on unit sphere
- Preserves unit length
- If $q_0 \cdot q_1 < 0$, negate one quaternion

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Combining Translation and Rotation

At time t:

$$\text{Position} = B(t)$$

$$\text{Orientation} = q(t)$$

- Translation and rotation are independent
- Both parameterized by same t

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Optional: Align Orientation to Path

Instead of SLERP:

- Use tangent:

$$\hat{v}(t) = \frac{B'(t)}{\|B'(t)\|}$$

- Construct quaternion that rotates forward axis to $\hat{v}(t)$
- Useful for cameras, vehicles, animation paths

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Summary

- Bézier curve provides smooth spatial path
- Quaternion SLERP provides smooth rotation
- Both avoid common issues:
  - No positional discontinuities
  - No gimbal lock

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